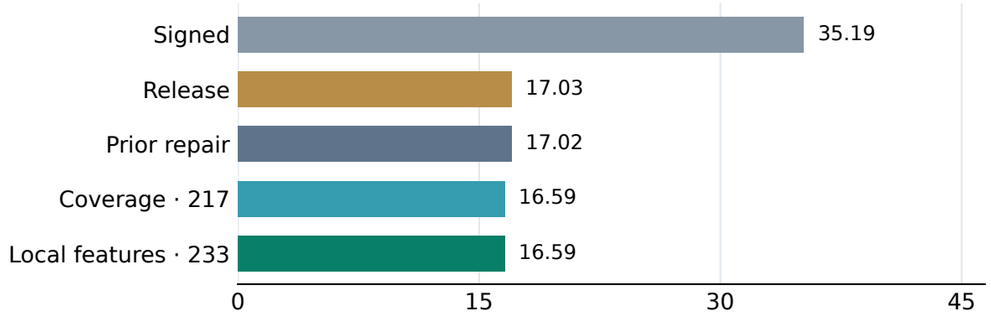
Training coverage and local features: matched comparison

Two fixed 300-tree fits · identical 1,857 training circuits · 32 new training labels · 217 vs 233 features

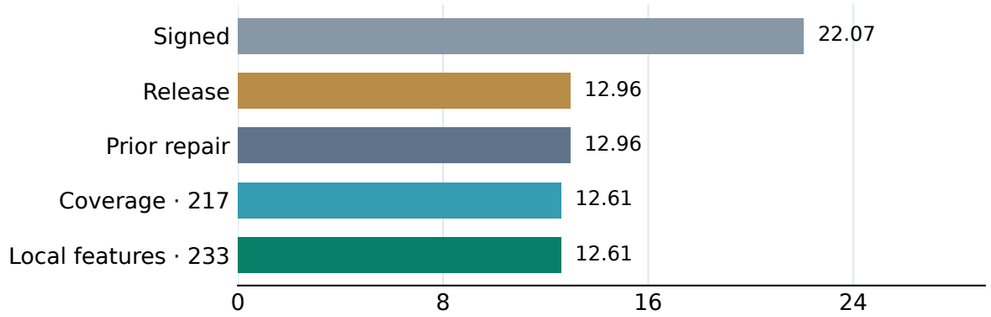
Development-selected policy: Coverage · 217 · Not eligible for promotion

General cohort · 32 fixed circuits

Policy MAE · LUT4 ↓



Policy positive-target MAPE · % ↓



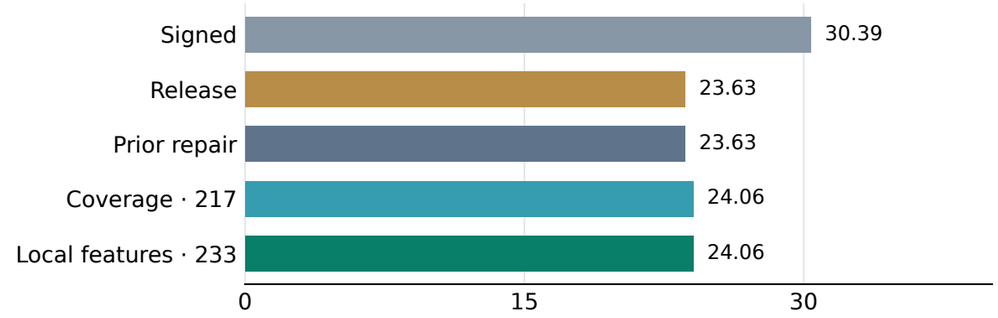
False accepts / false rejects · each budget assessed separately

Cap	Signed	Release	Prior	217	233
32	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
64	3 / 0	2 / 0	2 / 0	2 / 0	2 / 0
128	12 / 0	2 / 2	2 / 1	2 / 0	2 / 0

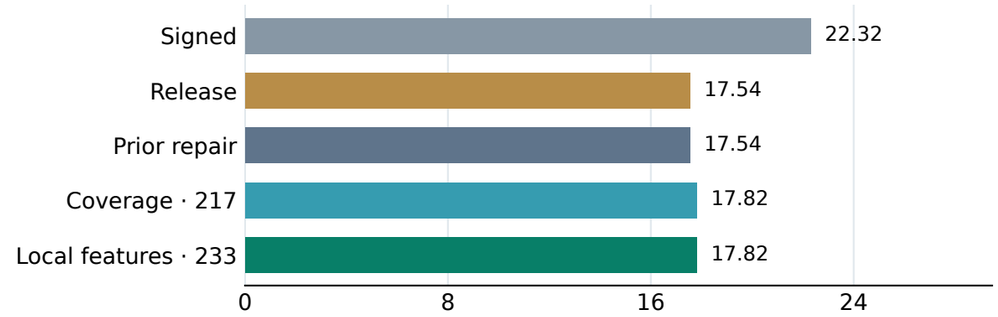
Feasible / infeasible: 32: 0/32 · 64: 4/28 · 128: 12/20

Conditional cohort · 1 active + 8 shared controls

Policy MAE · LUT4 ↓



Policy positive-target MAPE · % ↓



False accepts / false rejects · each budget assessed separately

Cap	Signed	Release	Prior	217	233
32	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
64	1 / 0	0 / 0	0 / 0	0 / 0	0 / 0
128	4 / 0	1 / 0	1 / 0	2 / 0	2 / 0

Feasible / infeasible: 32: 0/9 · 64: 2/7 · 128: 5/4

The new forests support a fixed cap-correction policy; bars show final policy predictions, not standalone raw-regressor accuracy.

Raw-regressor diagnostics and active-only gates are retained in the report. Candidate selection precedes fresh labels; no post-test model switch.

Eight controls are shared between cohorts. Conditional cases are selected by prediction disagreement; the cohorts are not pooled.

Measured target: isolated iCE40 LUT4 count. This is not a CPU speedup, application-transfer or universal hardware-safety result.